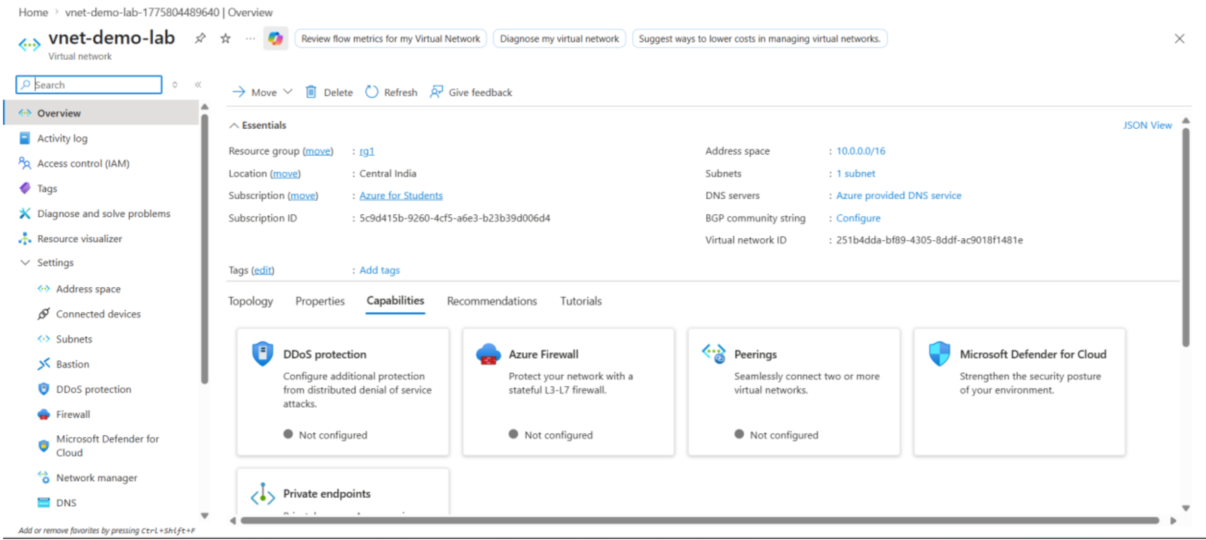
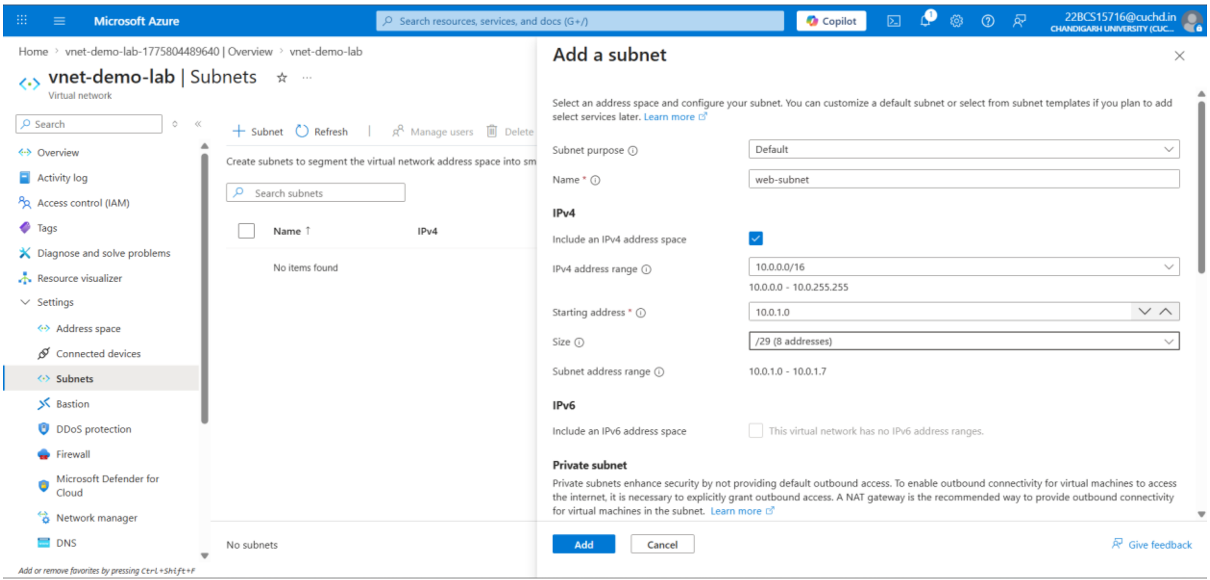
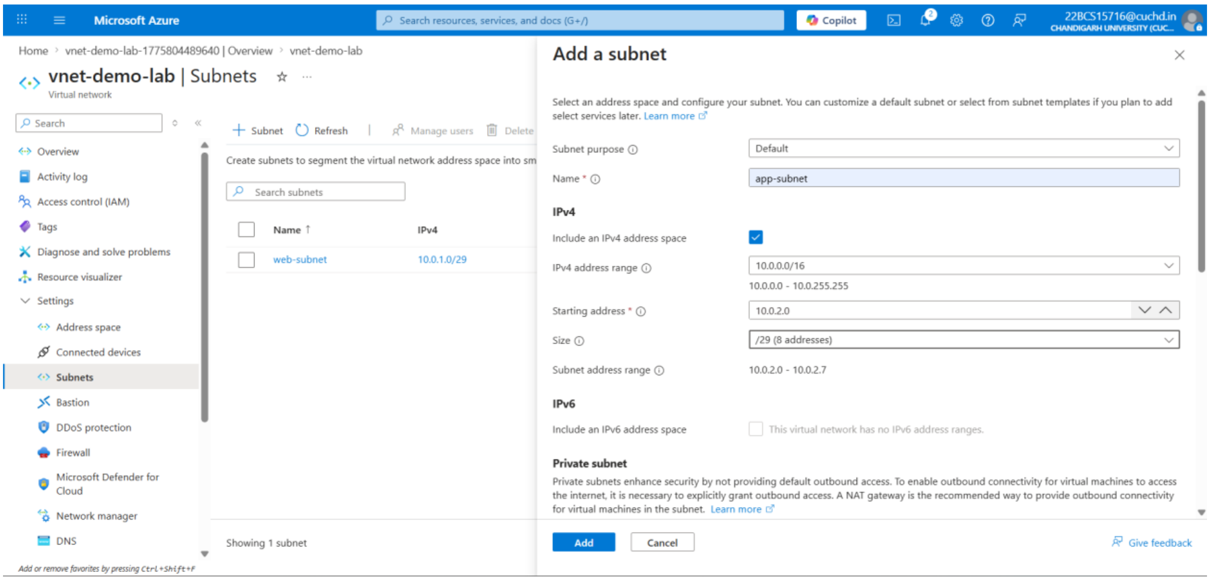
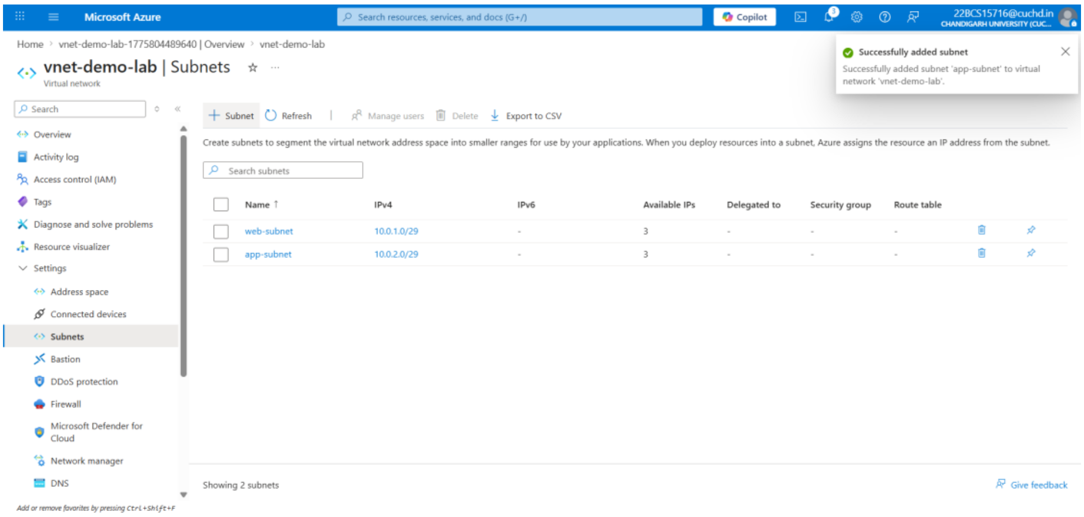
Azure VNet Exercise

Part 1: Create Virtual Network (Foundation)

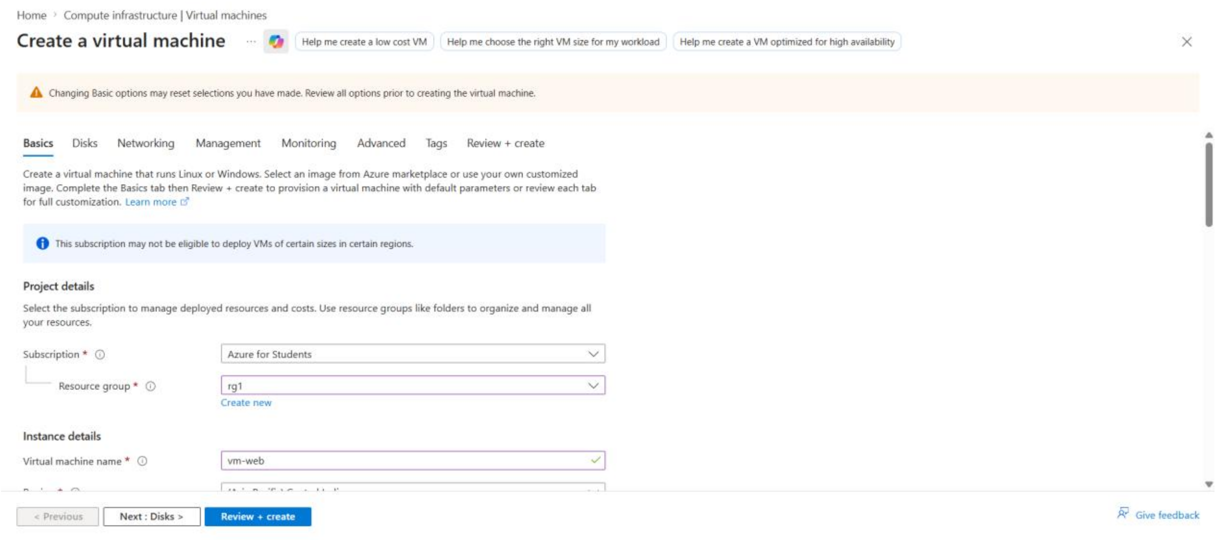


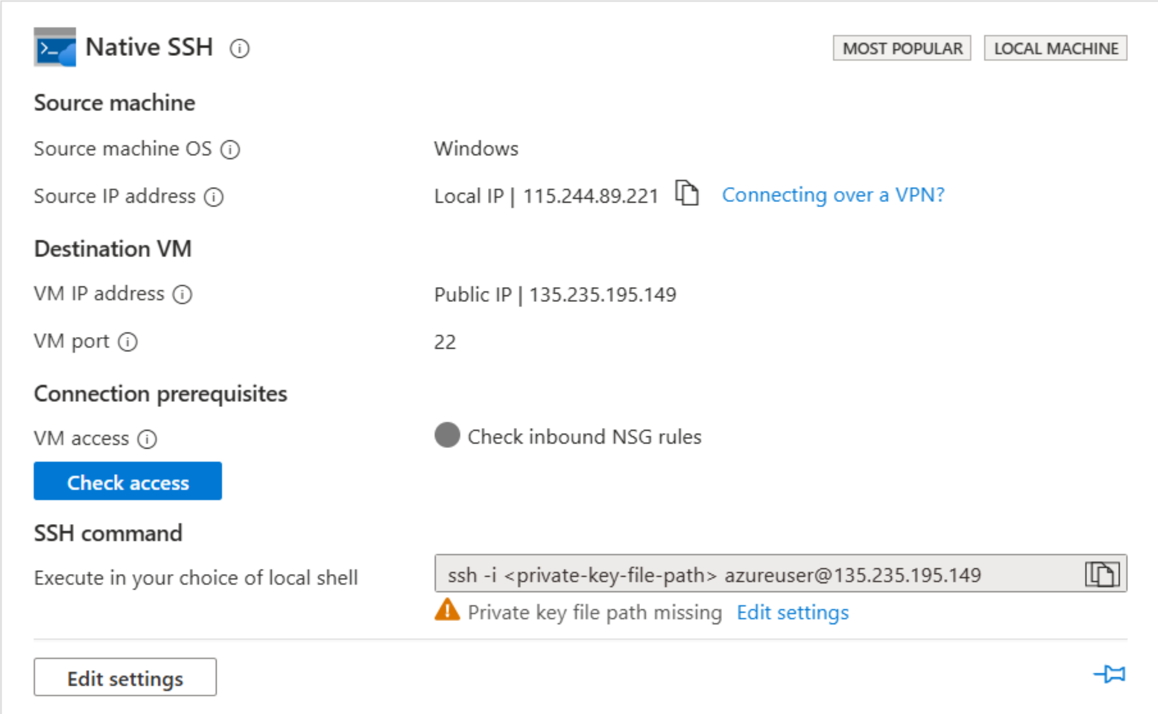


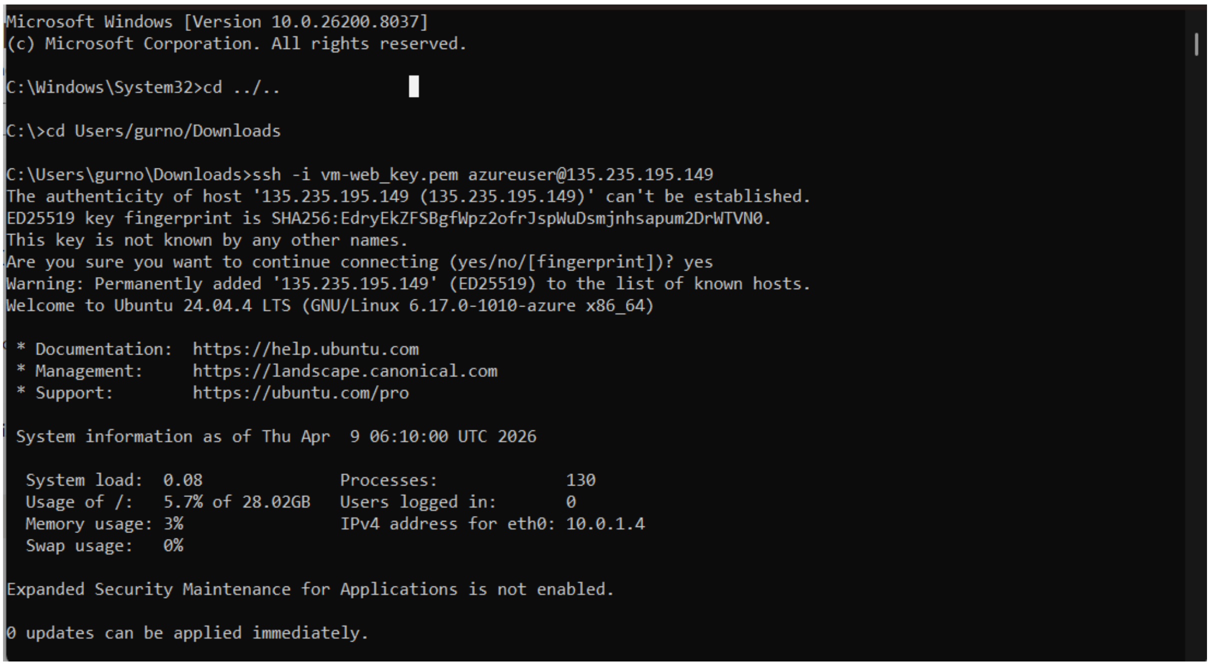




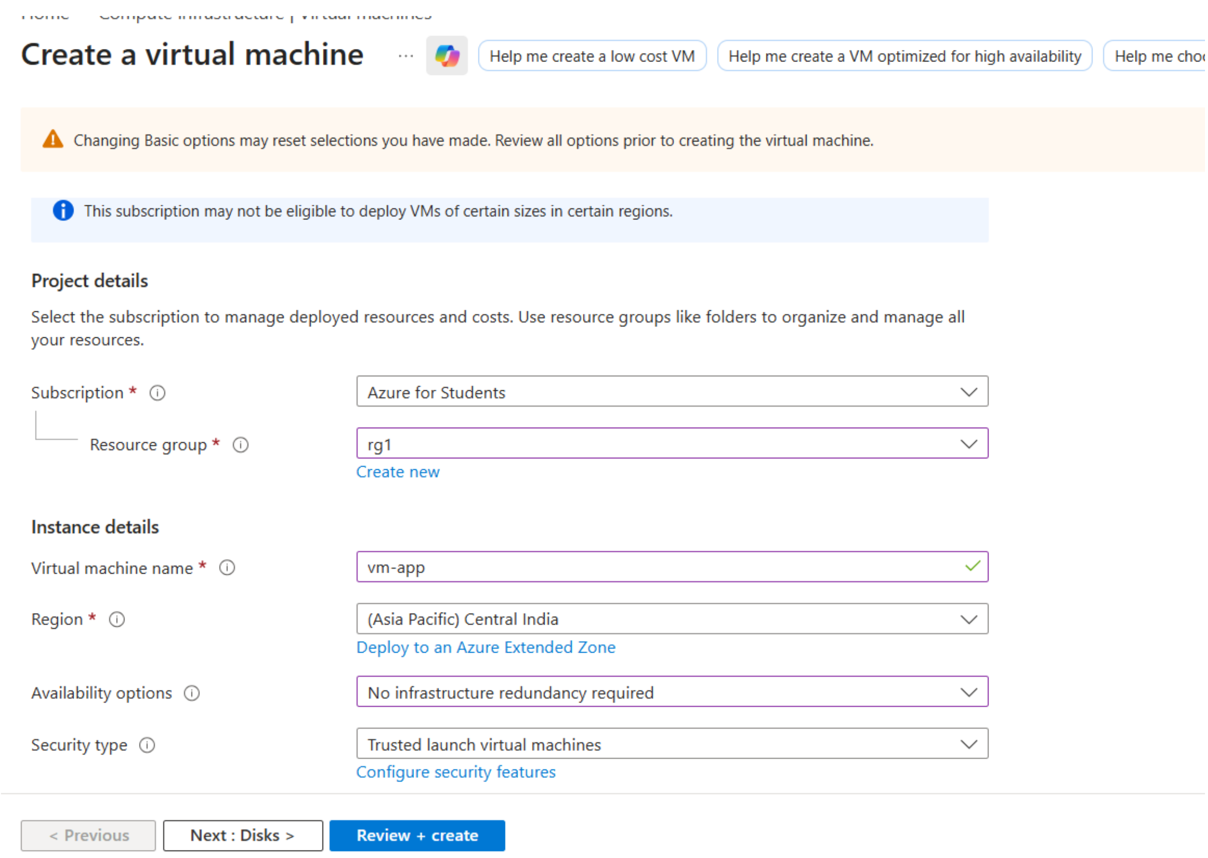
Part 2: Create Virtual Machines

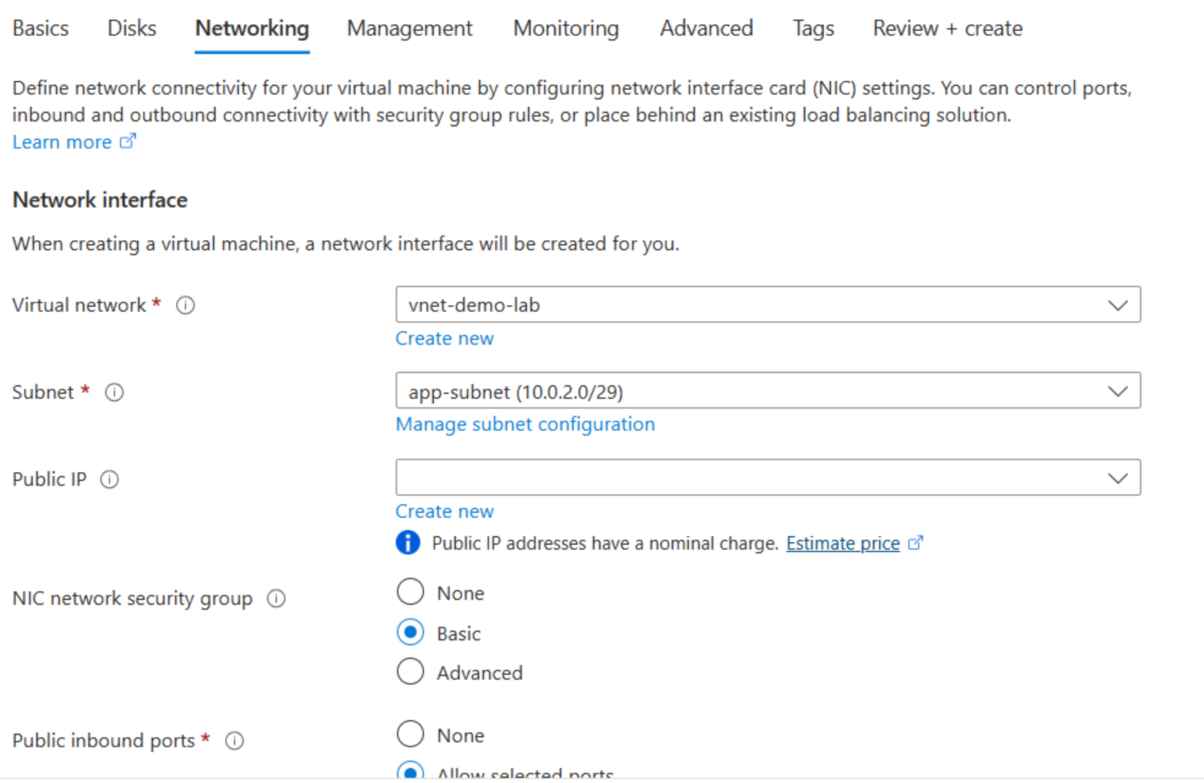


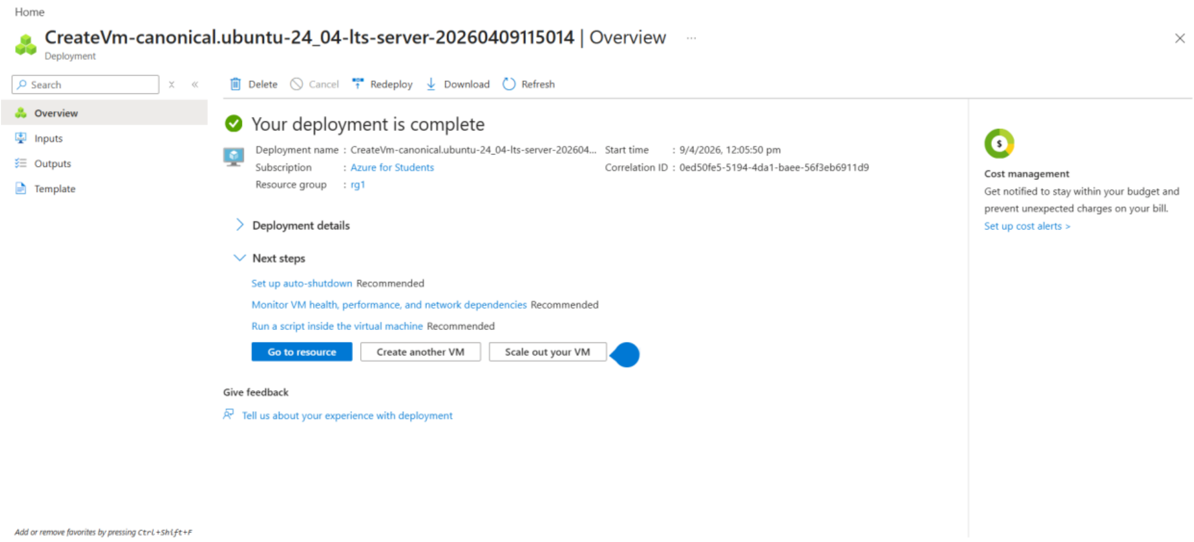


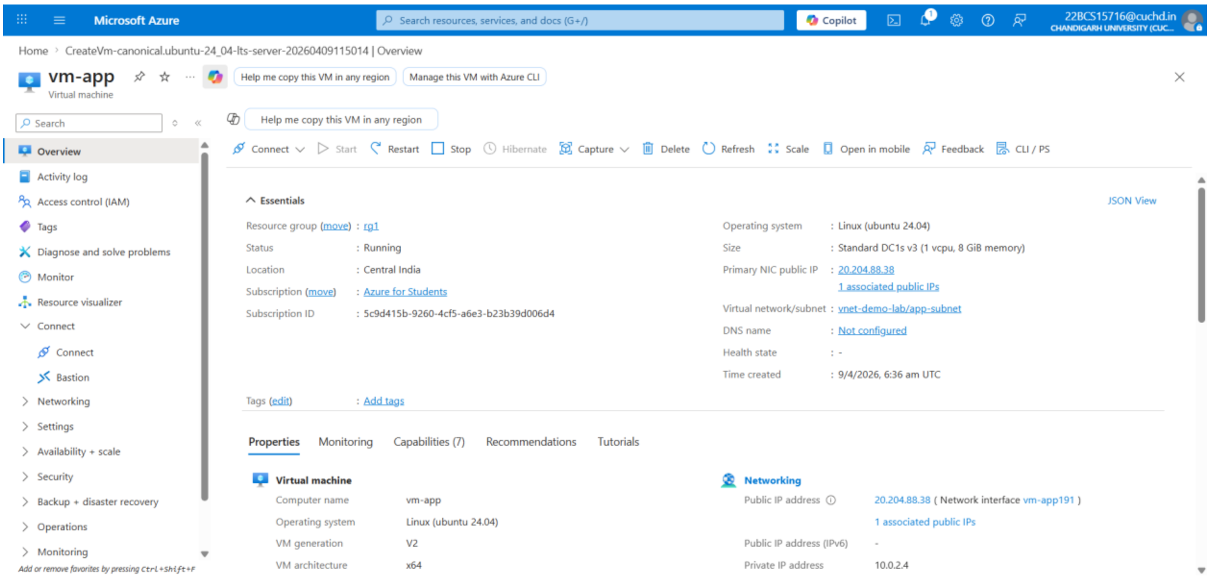


```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd ../..

C:\>cd Users/gurno/Downloads

C:\Users\gurno\Downloads>ssh -i vm-web_key.pem azureuser@135.235.195.149
The authenticity of host '135.235.195.149 (135.235.195.149)' can't be established.
ED25519 key fingerprint is SHA256:EdryEkZFSBgfWpz2ofrJspWuDsmjnhsapum2DrWTVN0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '135.235.195.149' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1010-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Apr  9 06:10:00 UTC 2026

  System load:  0.08              Processes:             130
  Usage of /:   5.7% of 28.02GB   Users logged in:       0
  Memory usage: 3%                IPv4 address for eth0: 10.0.1.4
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.
```

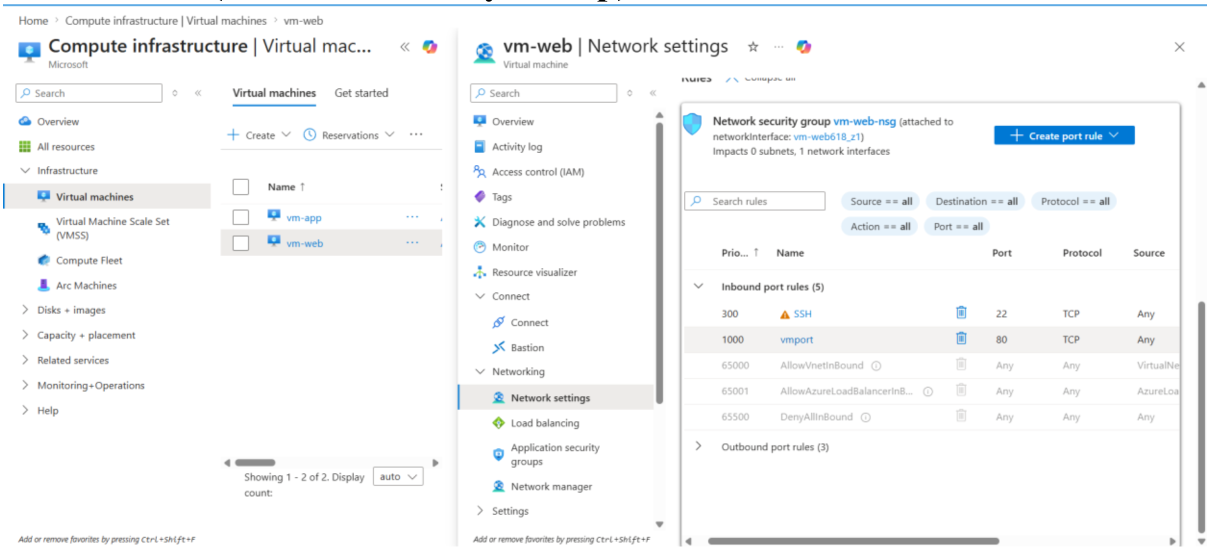

Create a virtual machine
Help me create a low cost VM
Help me create a VM optimized for high availability
Help me cho
Changing Basic options may reset selections you have made. Review all options prior to creating the virtual machine.
This subscription may not be eligible to deploy VMs of certain sizes in certain regions.
Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
Azure for Students
Resource group *
rg1
Create new
Instance details
Virtual machine name *
vm-app
Region *
(Asia Pacific) Central India
Deploy to an Azure Extended Zone
Availability options
No infrastructure redundancy required
Security type
Trusted launch virtual machines
Configure security features
< Previous
Next : Disks >
Review + create


Basics
Disks
Networking
Management
Monitoring
Advanced
Tags
Review + create
Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.
Learn more
Network interface
When creating a virtual machine, a network interface will be created for you.
Virtual network *
vnet-demo-lab
Create new
Subnet *
app-subnet (10.0.2.0/29)
Manage subnet configuration
Public IP
Create new
Public IP addresses have a nominal charge. Estimate price
NIC network security group
None
Basic
Advanced
Public inbound ports *
None





Part 3: NSG (Network Security Group)



```
azureuser@vm-web:~$ sudo ufw allow 'Nginx Full'
Rules updated
Rules updated (v6)
azureuser@vm-web:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
azureuser@vm-web:~$
```

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Part 4: Secure Network Traffic (Real Scenario)

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@vm-app:~$ ping 115.244.89.221
PING 115.244.89.221 (115.244.89.221) 56(84) bytes of data.
^C
--- 115.244.89.221 ping statistics ---
32 packets transmitted, 0 received, 100% packet loss, time 31782ms

azureuser@vm-app:~$ ping 10.0.2.4
PING 10.0.2.4 (10.0.2.4) 56(84) bytes of data.
64 bytes from 10.0.2.4: icmp_seq=1 ttl=64 time=0.019 ms
64 bytes from 10.0.2.4: icmp_seq=2 ttl=64 time=0.023 ms
64 bytes from 10.0.2.4: icmp_seq=3 ttl=64 time=0.018 ms
64 bytes from 10.0.2.4: icmp_seq=4 ttl=64 time=0.036 ms
64 bytes from 10.0.2.4: icmp_seq=5 ttl=64 time=0.034 ms
64 bytes from 10.0.2.4: icmp_seq=6 ttl=64 time=0.054 ms
64 bytes from 10.0.2.4: icmp_seq=7 ttl=64 time=0.055 ms
64 bytes from 10.0.2.4: icmp_seq=8 ttl=64 time=0.052 ms
64 bytes from 10.0.2.4: icmp_seq=9 ttl=64 time=0.051 ms
64 bytes from 10.0.2.4: icmp_seq=10 ttl=64 time=0.056 ms
^C
--- 10.0.2.4 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9245ms
rtt min/avg/max/mdev = 0.018/0.039/0.056/0.014 ms
azureuser@vm-app:~$
```